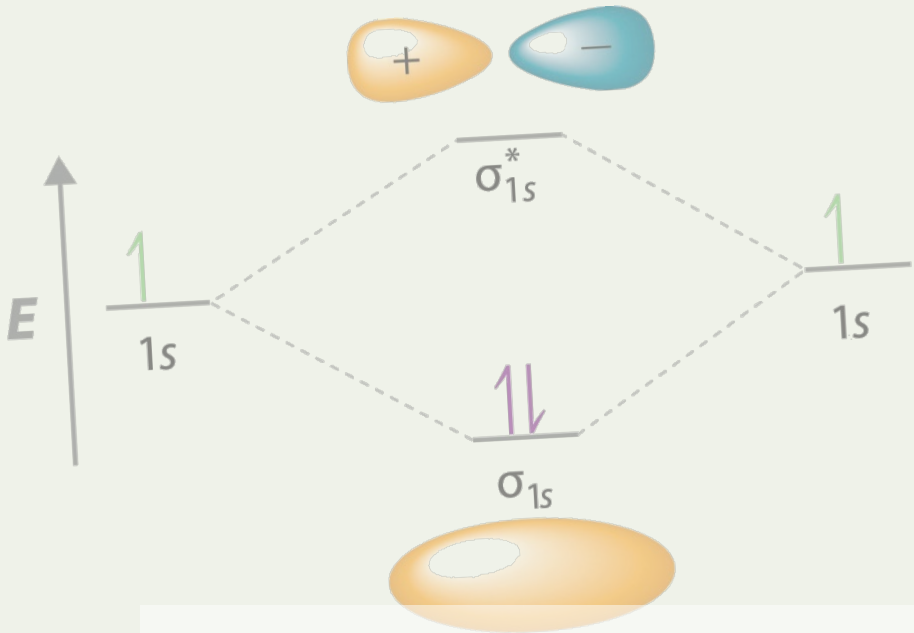


# Condensed Matter Theory

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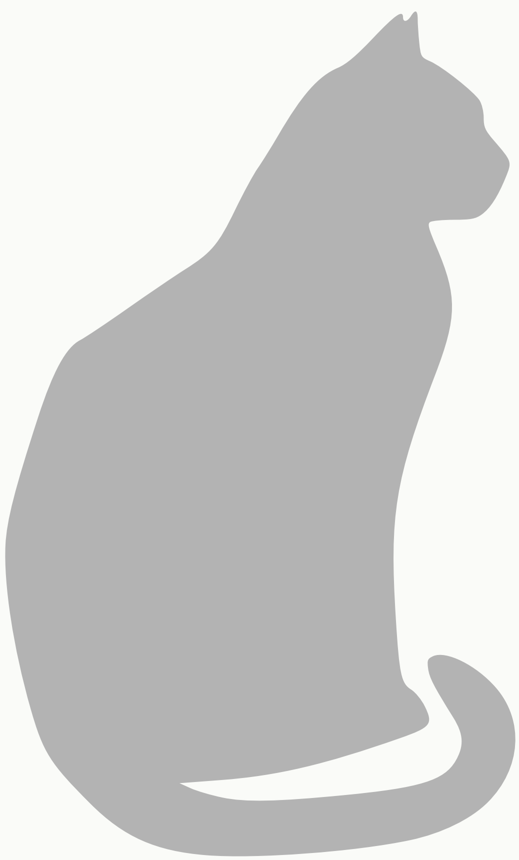

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# Contents

# Lecture #1 Electrons in a magnetic field

**Problem 1.1** (Onsager formula). Suppose that the electron spectrum for a tetragonal lattice can be described by the following equation

$$E_k = \frac{\hbar^2 k_x^2}{2m_{xx}} + \frac{\hbar^2 k_y^2}{2m_{xx}} + \frac{\hbar^2 k_z^2}{2m_{zz}},$$

with $m_{xx} \neq m_{zz}$. Using the Onsager formula for the cyclotron mass, $m_c = \frac{\hbar^2}{2\pi}\frac{\partial A}{\partial E_k}$ with $A$ being the area of an isoenergetic contour perpendicular to the magnetic field. Find $m_c$ for the case when the magnetic field is

(a) Along the $z$-axis.

(b) In the $x - y$ plane.

**Solution.**

(a) For $\boldsymbol{B} \parallel \hat{z}$, the motion of electrons will perpendicular to $\hat{z}$, i.e., parallel to $x - y$ plane. The equation of motion becomes

$$E_k = \frac{\hbar^2}{2m_{xx}}(k_x^2 + k_y^2),$$

with the equation of the motion boundary in $k$-space

$$k_\parallel^2 \equiv k_x^2 + k_y^2 = \frac{2m_{xx}E_k}{\hbar^2}.$$

Then, we get the radius. The circle area of the isoenergetic contour is

$$A = \pi k_\parallel^2 = \frac{2\pi m_{xx}E_k}{\hbar^2}.$$

Substituting to the definition of cyclotron mass, we can obtain

$$m_c = \frac{\hbar^2}{2\pi}\frac{\partial A}{\partial E_k} = \frac{\hbar^2}{2\pi}\frac{2\pi m_{xx}}{\hbar^2} = m_{xx}.$$

(b) Simply taking $\boldsymbol{B} \parallel \hat{x}$, then orbits lie in the $k_y - k_z$ plane with an ellipse area. The equation of motion becomes

$$E_k = \frac{\hbar^2 k_y^2}{2m_{xx}} + \frac{\hbar^2 k_z^2}{2m_{zz}},$$

with the equation of the motion boundary in $k$-space

$$\frac{k_y^2}{2m_{xx}E/\hbar^2} + \frac{k_z^2}{2m_{zz}E/\hbar^2} = 1.$$

Then we get the semi-axes. The ellipse area of the isoenergetic contour is

$$A = \pi\sqrt{\frac{2m_{xx}E}{\hbar^2}}\sqrt{\frac{2m_{zz}E}{\hbar^2}} = \frac{2\pi E}{\hbar^2}\sqrt{m_{xx}m_{zz}}$$

Substituting to the definition of cyclotron mass, we can obtain

$$m_c = \frac{\hbar^2}{2\pi}\frac{\partial A}{\partial E_k} = \sqrt{m_{xx}m_{zz}}.$$

**Problem 1.2** (Integer Quantum Hall Effect). The Hall resistance $R_{xy}$ of a two-dimensional electron gas is measured as a function of magnetic field $B$ at a very low temperature. A series of plateaus are observed at values $R_{xy} = \frac{h}{(\nu e^2)}$, where $\nu$ is an integer. Simultaneously, the longitudinal resistance $R_{xx}$ shows peaks and drops to zero.

(a) Explain physically why $R_{xx}$ drops to zero whenever $R_{xy}$ is on a plateau.

(b) A sample has an electron density of $n_s = 4.0 \times 10^{11}\,\mathrm{cm}^{-2}$.

  i. Calculate the filling factor $\nu$ at $B = 2.0\,\mathrm{T}$.

  ii. What is the expected value of $R_{xy}$ on the plateau nearest to this field?

  iii. How many Landau levels are completely filled at this plateau?

(c) Why is the IQHE used as a resistance standard? Which two fundamental constants are involved?

**Solution.**

(a) When the Hall resistance $R_{xy}$ is on a plateau, the Fermi energy lies in a mobility gap between Landau levels. All extended states in the bulk are absent, making the bulk insulating. Current flows only along dissipationless chiral edge states.

At the edges, backscattering is prevented by spatial separation: even if impurities or imperfections exist along an edge, they may cause small forward deviations in an electron's path but cannot reverse its direction, so no resistance is produced. With no extended states in the bulk to allow dissipation and no backscattering at the edges, the longitudinal voltage drop vanishes, giving $R_{xx} = 0$.

(b) i. From the degeneracy of Landau level $n_s = \nu \frac{eB}{h}$, we can obtain the filling factor

$$\nu = \frac{n_s h}{eB} = \frac{4.0 \times 10^{15}\,\mathrm{m}^{-2} \times 6.626 \times 10^{-34}\,\mathrm{J\cdot s}}{1.602 \times 10^{-19}\,\mathrm{C} \times 2.0\,\mathrm{T}} = 8.271.$$

  ii. Substituting the values into the expression of the Hall resistance

$$R_{xy} = \frac{R_K}{\tilde{\nu}} = \frac{h}{\tilde{\nu} e^2} = 3226.60\,\Omega.$$

  where $\tilde{\nu}$ is the nearest integer of $\nu$.

  iii. The filling factor $\tilde{\nu} = 8$ directly corresponds to the number of completely filled Landau levels.

(c) IQHE provides a quantized Hall resistance

$$R_H = \frac{h}{e^2},$$

which precisely quantized and only based on two fundamental constants: The Planck constant $h$ and the elementary charge $e$, so it is used as a resistance standard.

**Problem 1.3** (Lifshitz-Kosevich formula)**.** The Lifshitz-Kosevich formula describes the amplitude of the Shubnikov-de Haas (SdH) oscillations. The amplitude $\Delta R$ is proportional to the factor: $R_T = \frac{\lambda(T)}{\operatorname{sh}\lambda(T)}$, where $\lambda(T) = 2\pi^2 \frac{k_B T m^*}{\hbar e B}$.

(a) A researcher measures the SdH oscillation amplitude $\Delta R$ at a fixed magnetic field $B$ as a function of temperature $T$. How can they use this data to determine the effective mass $m^*$ of the charge carriers?

(b) Another researcher fixes the temperature $T$ and measures the SdH amplitude $\Delta R$ as a function of $1/B$. They find that the amplitude decreases exponentially with $1/B$. What physical parameter does this decay reveal, and what does a rapid decay imply about the quality of the sample?

**Solution.**

(a) Take $\alpha = \frac{2\pi^2 k_B m^*}{\hbar e B}$, then $\lambda(T) = \alpha T$. Since at $T \gg 1$, the amplitude $\Delta R$ satisfies

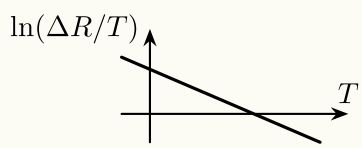
$$\Delta R \propto R_T = \frac{\alpha T}{\operatorname{sh}(\alpha T)} \to \frac{1}{2}\alpha T\, \mathrm{e}^{-\alpha T}, \quad \Delta R = \tilde{C} T\, \mathrm{e}^{-\alpha T}$$

Take the logarithm to the two sides

$$\ln \Delta R = \ln \tilde{C} + \ln T - \alpha T, \quad \ln\left(\frac{\Delta R}{T}\right) = -\alpha T + C,$$

where $C = \ln \tilde{C}$.

So, one can take the data at high temperature, and use some software tool to convert the original data from $T$-$\Delta R$ to $T$-$\ln\left(\frac{\Delta R}{T}\right)$, then plot the new data set like the right figure.




By fitting the slope $k = -\alpha$ that fits the figure, one can obtain the effective mass

$$m^* = \frac{\alpha \hbar e B}{2\pi^2 k_B} = -\frac{k \hbar e B}{2\pi^2 k_B}$$

(b) Similarly to part (a), we take $T \gg 1$. Since $\Delta R$ is proportional to $R_T R_D R_S$, and if we only consider the magnetic variable $B$, then $\Delta R \propto R_T R_D$. To be more specific,

$$\Delta R \propto \frac{2\pi^2 k_B T m^*}{\hbar e} \cdot \frac{1}{B} \cdot \mathrm{e}^{-\frac{2\pi^2 k_B T m^*}{\hbar e} \cdot \frac{1}{B}} \cdot \mathrm{e}^{-\frac{\pi m^*}{e \tau_q} \cdot \frac{1}{B}} = \tilde{C} \cdot \frac{1}{B} \cdot \mathrm{e}^{-(\alpha + \beta/\tau_q) \cdot \frac{1}{B}}$$

where the coefficients $\alpha$ and $\beta$ can be easily known since we have already obtained the effective mass. Taking the logarithm to both sides, we can obtain

$$\ln(\Delta R) = C - \ln B - \left(\alpha + \frac{\beta}{\tau_q}\right) \frac{1}{B},$$

Since the effective mass $m^*$ has been determined from (a), so the coefficients $\alpha$ and $\beta$ are known, and the slope directly reveals the quantum scattering rate $1/\tau_q$.

A rapid exponential decay corresponds to a short $\tau_q$, meaning a short mean free path and therefore significant disorder: an indicator of poor sample quality.

## Lecture #2 Band structure of graphene

**Problem 2.1.** In the calculation of the monolayer graphene band structure using tight-binding model, if we keep the contribution of the next-nearest hopping $t'$.

(a) Write down the dispersion relation.

(b) Expand the energy spectrum around the Dirac point up to second order. Search the literature and check what's trigonal warping effect is in a monolayer.

**Solution.**

(a)

**Problem 2.2.** Graphene has the record room-temperature electron mobility, mainly due to two reasons. The first is the high Debye temperature ($\sim 2100\,\mathrm{K}$) that suppresses phonon scattering arising from the in-plane covalent $sp^2$ bonding between carbon atoms. The second one is related to its unique linear Dirac band, resulting in massless Dirac fermions ($m^* = 0$) near the Dirac point. However, if we apply the definition of the effective mass $m^* = \hbar^2 \left(\frac{\mathrm{d}^2 E(k)}{\mathrm{d}k^2}\right)^{-1}$ to the band dispersion of graphene ($E_\pm(k) = \pm\hbar v_F k$), we will get diverging $m^*$ and non-differentiable Dirac point. Please explain this contradiction.

**Solution.** The contradiction arises from applying different definitions of "mass" to graphene's unique band structure. The description of "massless Dirac fermions" comes from the analogy to relativistic particles.

From a massless special relativity particle like a photon to the electrons in the graphene, their dispersion relations have the comparasion

$$E = pc \Longrightarrow E = \pm\hbar v_F k = p v_F$$

with $v_F$ playing the role of $c$. Hence, electrons near the Dirac point behave as if they have zero rest mass.

However, the conventional effective mass $m^* = \hbar^2(\mathrm{d}^2E/\mathrm{d}k^2)^{-1}$ is defined under the parabolic-band approximation $E \propto k^2$. For graphene's linear dispersion, $\mathrm{d}^2E/\mathrm{d}k^2 = 0$ away from $k = 0$, leading to $m^* \to \infty$ mathematically. At $k = 0$ the dispersion is non-differentiable, so the formula breaks down.

Physically, this infinite effective mass reflects that the electron's speed is constant at $v_F$: An external force parallel to its momentum cannot change its speed, much like a photon's speed is fixed at $c$. Thus, the "massless" label refers to the relativistic analogy, while the divergent $m^*$ indicates the inapplicability of the parabolic effective mass concept to a linear band.